

Endogenous Population Scale in the Existing Housing–Fertility Model

Research update

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Abstract

The existing lifecycle model determines tenure, housing, and fertility while normalizing adult population. This update keeps that household problem intact and lets births affect the number of future households. A simple benchmark shows that replacement fertility and housing clearing jointly determine the long-run housing cost and population scale. The maintained calibration can be embedded in an open stationary equilibrium with fixed outside entry, and the active solver now implements that scale equation in its existing price loop. It cannot yet be interpreted as an observed point on a transition between closed steady states: its fertility response to housing costs is too weak to produce a positive closed endpoint, and its normalized stationary distribution does not contain the observed U.S. cohort structure. The quantitative result at this stage is therefore an incidence comparison, not a calibrated U.S. transition.

1 The mechanism

Children raise the demand for space at the same ages when households have low liquid wealth and limited access to ownership. Large rental units are scarce, and buying a family-sized home requires a down payment. Older households, by contrast, own much of the large-housing stock and may retain it after their children leave because moving is costly and housing is part of the estate they plan to leave. The allocation of housing across generations can therefore raise the effective cost of a child even when the aggregate stock is large.

The feedback has two directions. First, a higher housing cost or a tighter access constraint reduces fertility. Second, children eventually form the next generation of households, so fertility changes future housing demand. The first direction is a household mechanism; the second is a demographic equilibrium condition. Combining them changes both the transition and the long-run incidence of housing policy.

Neither direction of the feedback is new. Demographic effects on housing demand go back at least to Mankiw and Weil (1989); recent evidence shows that birth cohorts predict owner-occupied housing prices decades later through age-concentrated entry into and exit from homeownership (Franke and Korevaar, 2025). Housing prices and mortgage access affect childbearing, with different exposure for renters, prospective buyers, and incumbent owners (Lovenheim and Mumford, 2013; Dettling and Kearney, 2014; Hacamo, 2021). Malmberg (2012) documents the broader two-sided association between cohort housing pressure and fertility, and Lee and Suh (2025) study a low-fertility housing transition in an OLG model. The contribution cannot therefore be that demography affects housing, or even that the feedback can be slow. The narrower candidate contribution is policy incidence in a model where tenure, constrained access, old-age retention, and fertility are household choices.

The paper-facing modification is one stationary wrapper around the existing model. At a candidate housing cost p , the maintained solver produces a normalized composition g , required young entrants $E_0(p, g)$, mature locally born entrants $B_0(p, g)$, and housing demand per adult $h(p, g)$. Fixed outside

entry M and local retention ρ close scale and the housing market through

$$SE_0(p, g) = M + \rho SB_0(p, g), \quad S\bar{h}(p, g) = H^s(p). \quad (1)$$

The entrant and maturation flows already existed in the solver. The renewal routine also existed, but the active Markov-income price-clearing branch did not call it; this update adds that missing call. The exercise holds the household block and the closure parameters fixed and solves these two scalar conditions jointly. It then reports fertility per household, population scale, and total births. No new household state, Bellman equation, or stationary distribution algorithm is required.

This scalar closure assumes that outside and locally born entrants draw the same fixed entrant-state distribution used by the maintained solver. It is therefore exact for the model's present aggregate renewal accounting, but it does not yet endogenize the wealth or tenure composition of descendants.

Section 2 makes the logic transparent with young and old households. Section 3 reports what works in the maintained calibration and what remains necessary before calling it a U.S. transition.

2 Two-generation theory

2.1 Environment

Young households. At date t , a mass Y_t of young households has income y . A young household chooses consumption c , adult housing h , and children n . Preferences are

$$u(c, h, n) = c + \alpha \log h + \theta \log n, \quad \alpha > 0, \quad \theta > 0. \quad (2)$$

The parameter θ is the value of children. Quasi-linearity removes income effects and keeps the price mechanism available in closed form. Throughout this section, income is high enough that the interior allocation is feasible: $y > \alpha + \theta$.

Child costs and young access. Each child requires $q \geq 0$ units of ordinary goods and $a > 0$ units of housing. The market cost of that housing is ap , where p is the price of housing services. A goods-equivalent wedge $\omega \geq 0$ captures the additional cost of obtaining family space through a small rental market, a down payment, or another tenure constraint. The full cost of a child is therefore $q + a(p + \omega)$.

Old households. A mass O_t of old households no longer chooses fertility. After solving its retention and estate problem, an old household's housing demand is summarized by

$$d_O(p; \tau) = \frac{\varphi(\tau)}{p}, \quad \varphi'(\tau) > 0. \quad (3)$$

The parameter τ indexes forces that keep old households in housing, including transaction costs and the estate motive. The lifecycle model below derives retention from choices rather than imposing equation (3).

Housing. Aggregate housing services are supplied according to

$$H^s(p) = \bar{H}p^\varepsilon, \quad \bar{H} > 0, \quad \varepsilon \geq 0. \quad (4)$$

The fixed-stock case is $\varepsilon = 0$. The number $\bar{n} > 0$ is effective replacement fertility: after survival and household formation, a young household with $\bar{n}$ children produces one young household in the next generation.

In this static section, p is the current cost of one unit of housing services. The lifecycle model distinguishes that current rent, denoted r_t , from the asset price P_t that capitalizes future rents. They move proportionally only under a constant user cost and a stationary price path.

2.2 The young household

Taking p as given, a young household solves

$$\max_{c,h,n} c + \alpha \log h + \theta \log n \quad \text{subject to} \quad c + ph + [q + a(p + \omega)]n = y. \quad (5)$$

At an interior solution, child demand and total housing demand are

$$n_Y(p; \theta, \omega) = \frac{\theta}{q + a(p + \omega)}, \quad d_Y(p; \theta, \omega) = \frac{\alpha}{p} + an_Y(p; \theta, \omega). \quad (6)$$

A higher market price and a tighter access wedge have the same sign in the child decision. They differ in equilibrium incidence: the market price clears housing, while the access wedge changes which households can obtain it.

2.3 Equilibrium and the steady state

Housing clearing adds young and old demand:

$$\bar{H}p_t^\varepsilon = Y_t d_Y(p_t; \theta, \omega) + O_t d_O(p_t; \tau). \quad (7)$$

For inherited cohort masses, the left side rises and the right side falls with p_t , so the clearing price is unique.

Children become the next young cohort, while the current young become old:

$$Y_{t+1} = \mathcal{R}(p_t; \theta, \omega)Y_t, \quad O_{t+1} = Y_t, \quad \mathcal{R}(p; \theta, \omega) = \frac{n_Y(p; \theta, \omega)}{\bar{n}}. \quad (8)$$

The function $\mathcal{R}$ is effective reproduction. It is strictly decreasing in the housing cost and the young-access wedge.

Definition 1 (Reproductive steady state). A reproductive steady state is a price $p^* > 0$, positive cohort masses (Y^*, O^*) , and household choices such that households optimize, housing clears, and $(Y_{t+1}, O_{t+1}) = (Y_t, O_t)$.

Proposition 1 (Steady state and policy incidence). *A unique positive reproductive steady state exists if and only if $\theta > \bar{n}(q + a\omega)$. It is*

$$p^* = \frac{\theta/\bar{n} - q}{a} - \omega, \quad Y^* = O^* = \frac{\bar{H}(p^*)^{\varepsilon+1}}{\alpha + \varphi(\tau) + a\bar{n}p^*}. \quad (9)$$

A lower value of children θ or a tighter young-access wedge ω lowers both the long-run housing cost and household population. Stronger old retention raises $\varphi(\tau)$ and lowers population without changing the reproductive housing cost. A proportional increase in housing supply $\bar{H}$ raises population in the same proportion and leaves the reproductive housing cost unchanged.

Reproduction and housing clearing perform different jobs. Replacement selects the housing cost at which a cohort reproduces itself. Housing clearing then selects how many young and old households can occupy the market at that cost. This triangular ordering is why permanent housing supply is capitalized into population rather than the long-run price in the closed economy.

The existence condition is economically sharp. Even if market housing were free, a child would still cost $q + a\omega$. If $\theta \leq \bar{n}(q + a\omega)$, cheaper market housing alone cannot restore replacement.

The same separation changes how a permanent housing policy should be read. Start from a reproductive steady state and let an unexpected reform arrive after cohort sizes are inherited. Write $g_Y = -\omega$ for easier young access, $g_O = -\tau$ for lower old retention, and $g_H = \log \bar{H}$ for greater housing supply, so an increase in any g is the stated reform.

Corollary 1 (Demographic capitalization). *For a local reform at an interior steady state, the impact and long-run signs are*

	Impact p_t	Impact n_t	Long-run p^*	Long-run n^*	Long-run population
Easier young access, g_Y	+	+	+	0	+
Lower old retention, g_O	−	+	0	0	+
Greater housing supply, g_H	−	+	0	0	+

The access result is not a paradox. The reform lowers the non-price cost of family housing, raises fertility and demand, and is partly capitalized into the market price. Thus a higher observed price need not mean that young households are worse off. Old release and new supply instead lower the price on impact. At any new reproductive steady state, demographic expansion has absorbed the released stock: effective fertility is back at replacement, the reproductive housing cost is at its original level, and the persistent effect is a larger population. A fixed-population comparison misses this change in incidence.

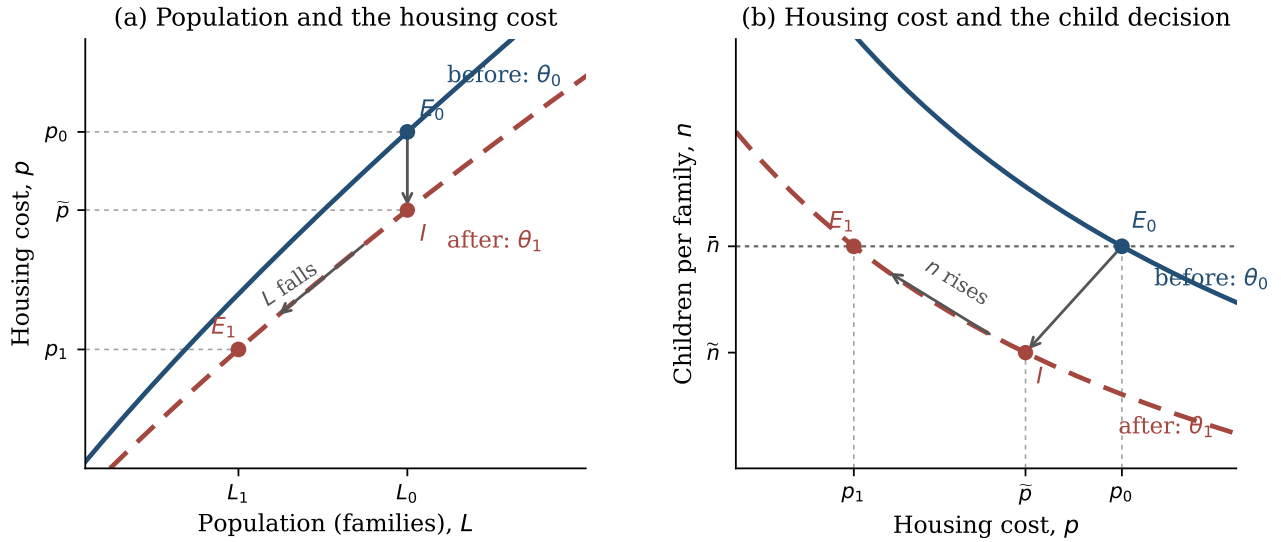


Figure 1: Reproduction selects the steady-state housing cost, and housing clearing maps that cost into population. After a fall in the value of children, fertility is initially below replacement. A lower population then reduces the price and partly offsets the original shock.

2.4 Demographic momentum

The steady-state result does not imply that population and prices must already be falling. This is a simple accounting point, not a separate theorem. Let $\mu_t = O_t/Y_t$ be the old-to-young ratio and let $\mathcal{R}_t = \mathcal{R}(p_t; \theta, \omega)$ be current effective reproduction. When $\mathcal{R}_t < 1$, household population still rises between t and $t + 1$ if and only if

$$\mu_t < \mathcal{R}_t < 1. \quad (10)$$

For example, 100 young and 70 old households become 80 young and 100 old households when $\mathcal{R}_t = 0.8$: population rises from 170 to 180 despite below-replacement fertility. The housing cost rises if and only if

$$(1 - \mu_t)d_O(p_t; \tau) > (1 - \mathcal{R}_t)d_Y(p_t; \theta, \omega). \quad (11)$$

Thus low fertility, population growth, and housing-cost growth coexist when

$$\mu_t < \min \left\{ \mathcal{R}_t, 1 - (1 - \mathcal{R}_t) \frac{d_Y(p_t; \theta, \omega)}{d_O(p_t; \tau)} \right\}. \quad (12)$$

Equation (11) is an accounting test. The smaller next young cohort removes $(1 - \mathcal{R}_t)Y_t d_Y$ units of demand. The unusually large current young cohort adds $(Y_t - O_t)d_O$ units as it ages. The housing cost rises when the second force is larger. From an exact stationary age distribution, $\mu_t = 1$, so both population and the housing cost fall after a fertility decline. A positive transition requires an inherited cohort bulge generated by earlier fertility, immigration, or both.

The fertility decline nevertheless lowers housing costs relative to the no-shock path. Demographic momentum can make the realized cost rise over time; it cannot reverse the counterfactual effect of the shock.

2.5 Intergenerational policy

Proposition 1 distinguishes three interventions. Relaxing the young-access wedge ω raises fertility at a given cost and supports a higher reproductive housing cost and population. Reducing old retention τ releases housing, lowers the transition cost, and raises fertility during adjustment; in the new closed-economy steady state it raises population while the reproductive housing cost returns to the level selected by θ and ω . Expanding housing supply has the same long-run housing-cost neutrality but does not directly change which generation occupies the stock.

This is where the demographic closure strengthens the paper’s original intergenerational mechanism. Reallocating one home toward a constrained young household can affect its current tenure and fertility, and that birth changes the size of the future entrant cohort. The second effect is absent when the adult population is normalized after every solution.

3 Mapping to the maintained quantitative model

3.1 What is retained and what is new

The immediate paper exercise retains the E5b household problem and shared child clock exactly. Children shift equivalence-scaled needs toward housing; fertility is sequential; renters face a size cap; buyers face the retained down-payment constraint; owners choose from the existing housing ladder; and transaction costs and warm-glow bequests generate old-age retention. The normalized stationary solver already reports `entry_rate`, `entrants_mature_total`, adult mass, and housing demand. Its renewal-scale routine evaluates equation (1), but the active Markov-income equilibrium branch previously ignored the resulting scale when clearing the housing market. The minimal code change is to use scaled demand in that existing price loop, attach the renewal residual to the accepted solution, and leave every household object unchanged. The normalized and renewal baselines now nest to within 5×10^{-7} in the asset price, with stationary scale equal to one up to 3.1×10^{-8} .

No child-clock change is part of this update. A calendar-time transition would add a level distribution, the existing child-maturation pipeline in levels, and time-indexed prices. Independent child clocks remain a separate model revision that would require re-estimation.

The intergenerational emphasis is strengthened, not displaced. Young-access policies have an immediate tenure and fertility effect and a delayed population effect. Old-release policies have an immediate stock-allocation effect and an indirect fertility effect through prices. Reporting both components reveals which policies change family formation and which chiefly reallocate the existing housing stock.

The two-generation model is therefore exposition for the existing paper, not a replacement model. The quantitative update defines (E_0, B_0, M, ρ, S) and reports how the paper’s policies change fertility per household, stationary scale, prices, and total births.

3.2 Audited benchmark

The best retained July 25 E5b parameter vector has current-contract loss 385.8755, asset price 0.815625, and housing residual 1.11×10^{-6} . It is an audited benchmark, not a promoted calibration: the PSID rooms construction is under hold, and childlessness and old-wealth dispersion remain large failures. Table 1 reports a few economically central rows.

Table 1: Selected audited fit of the retained benchmark

Moment	Data	Model	Reading
Completed fertility	1.918	1.904	Close
Family ownership gap	0.168	0.166	Close
Rooms gap, 3+ versus 1-2 children	0.368	0.372	Close
Ownership, ages 30-55	0.575	0.564	Close
Childless share	0.188	0.080	Large miss
Old wealth p90/p50	3.448	2.068	Large miss

The benchmark requires entrant flow $E_0 = 0.0617335$ per adult and produces mature locally born flow $B_0 = 0.0511056$. Thus

$$\frac{B_0}{E_0} = 0.82784 < 1. \quad (13)$$

Treating the 3+ category with its reported top-code mean raises the diagnostic ratio to 0.90319, still below replacement. Across asset prices from 0.01 to two times the benchmark, the maintained ratio ranges only from about 0.836 to 0.820; its local price elasticity is -0.01067 . The maintained benchmark therefore has no positive closed reproductive root over the audited range.

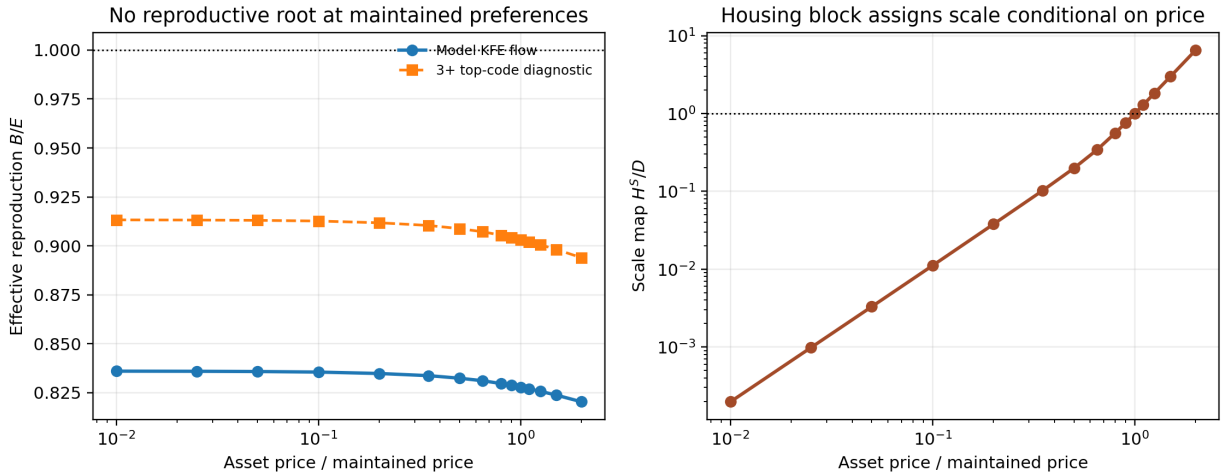


Figure 2: Reproductive and housing-scale maps at the retained benchmark. The left panel shows that neither the model flow nor the 3+ top-code diagnostic reaches replacement over the audited price range. The right panel shows that housing clearing nevertheless assigns stationary scale conditional on price.

3.3 What the open closure delivers now

Keep every locally born mature child and choose the fixed outside flow so that the benchmark has scale one. Then $\rho = 1$ and

$$M = E_0 - B_0 = 0.010628, \quad \frac{M}{E_0} = 0.17216. \quad (14)$$

Thus outside entry must supply 17.2 percent of required entrants, or 1.06 percent of the adult-household stock every four years. This is an accounting normalization, not yet an empirical migration estimate: the model compresses entry at different ages into one entrant state.

Table 2 reruns the retained shared-clock household model under this fixed-inflow closure. The first price column is the old normalized comparison; the second lets stationary population scale affect housing demand.

Table 2: Legacy policy cases with endogenous stationary scale

	Δ TFR/HH (%)	Δp , normalized (%)	Δp , renewal (%)	ΔS (%)	Δ total births (%)
Birth grant for family-size homes	0.082	0.458	0.599	0.378	0.458
Property tax plus birth grant	0.126	−18.896	−18.714	0.579	0.700

The closure adds a demographic margin, but it does not overturn these policy results. The grant supports a population about 0.4 percent larger; the tax-and-grant case supports one about 0.6 percent larger. The reason is quantitative rather than conceptual: even a large housing-price decline barely moves effective reproduction in the maintained calibration.

The local incidence calculation makes the contrast sharp. At the normalization in equation (14), a one-percent direct increase in mature locally born entrants raises stationary population by about 4.72 percent before accounting for any direct housing-demand shift. A one-percent housing supply expansion, by contrast, lowers the price by about 0.37 percent but raises population by only 0.02 percent, because cheaper housing currently has almost no effect on reproduction. These are diagnostic elasticities, not estimated policy effects.

3.4 Can the calibration be a point on a transition?

Not yet in the strong sense. With the normalized stationary distribution and the fixed flow in equation (14), the maintained point is an *open steady state*; absent another shock, nothing moves. Relabeling that distribution as “today” would not create demographic momentum.

A credible transition exercise can nevertheless use the estimated household block as its behavioral component. It must replace the normalized age profile with an observed initial distribution of household heads, carry the children already in the maturation pipeline, specify migration by age, and solve the sequence of housing costs. We can start from today’s measured state without claiming to observe the historical preference shock. The terminal point under the maintained calibration should be open, because no positive closed root has been found.

This gives a clean separation for the paper. The stationary open closure is a small, feasible update that can be reported now. A path from today’s cohort distribution to that endpoint is a second quantitative exercise. A transition between two positive *closed* steady states would additionally require a fertility block whose response to housing costs is strong enough to restore replacement; the maintained calibration does not have that property.

3.5 Identification consequence

Stationary fertility levels and timing observed at one market price do not identify the slope R_{0P} . The current target system contains strong moments for levels, timing, tenure, and rooms, but no causal housing-cost-to-birth hazard moment. The near-flat reproductive map in Figure 2 could therefore reflect either the model’s true economic restriction or an unidentified fertility-noise scale.

A quantitative transition needs a tenure-specific response moment that moves housing cost or access

while holding other child costs fixed. The estimand should distinguish renters, prospective buyers, and incumbent owners because the same asset-price movement raises rent and down payments but also raises owner wealth. Without this slope evidence, the steady-state feedback and policy amplification should be presented as theory, not as an estimated magnitude.

4 Conclusion

The proposed paper update is small: retain the household model, stop normalizing scale in stationary policy comparisons, and clear housing using the resulting population. The simple model explains the endpoint logic; the retained quantitative model now executes the open version of that closure.

The first numerical result is disciplined but modest. Endogenous scale adds a population response of roughly 0.4–0.6 percent to the legacy grant cases and slightly changes their price incidence; it does not reverse their conclusions. The maintained fertility schedule is too insensitive to housing costs for supply or price reductions to generate large demographic feedback.

The more ambitious between-steady-states interpretation remains a hypothesis, not a completed result. Today can be used as an initial condition only after the normalized KFE is replaced by the observed cohort distribution and child pipeline. With the current fertility block, the relevant terminal equilibrium is open rather than a positive closed steady state. This is the focused next step if the advisor views the transition as central to the paper.

A Benchmark audit tables

Table 3: Retained E5b benchmark: complete current-contract target system

Moment	Target	Model	Gap	Weight	Loss contribution
Completed fertility	1.918000	1.903573	−0.014427	1425.739	0.297
Childless share	0.188000	0.080188	−0.107812	17180.744	199.700
Mean age at first birth	25.310561	25.637433	0.326872	16.000	1.710
Share of first births at age 30 or later	0.270062	0.332334	0.062272	15625.000	60.590
Rooms response at first birth (measurement hold)	0.804944	0.657205	−0.147739	35.735	0.780
Rooms gap, 3+ versus 1–2 children	0.367700	0.372240	0.004540	2958.515	0.061
Family ownership gap	0.167662	0.165878	−0.001783	14229.591	0.045
Ownership rate, ages 30–55	0.575472	0.563758	−0.011714	1207.846	0.166
Mean occupied rooms, ages 18–85	5.779970	6.075505	0.295534	11.973	1.046
Wealth / annual gross labor earnings	6.873100	5.454754	−1.418346	6.288	12.649
Annual bequest flow / aggregate wealth	0.008800	0.009078	0.000278	5165289.256	0.399
Old wealth-to-income p90/p50, ages 76–84	3.448111	2.068370	−1.379741	56.960	108.433
Exact current-contract loss					385.875

The table is an audit artifact, not a promoted calibration table. In particular, the first-birth rooms row remains under a PSID measurement hold. The unrounded row contributions sum exactly to the reported scalar; the small difference visible after summing the printed entries is rounding.

Table 4: Retained E5b benchmark: all free and externally fixed parameters

Parameter	Estimate	Search bound / restriction	Status
Annual discount factor β	0.991097	[0.94, 0.9995]	Estimated
Per-child expenditure shift Δ_1	0.012163	[0, 0.25]	Estimated
First-child expenditure jump Δ_0	0.023896	[0, 0.25]	Estimated
Value-of-child shifter ψ	0.585018	[-3, 3]	Estimated
First-birth logit scale κ_1	3.552006	[0.02, 50]	Estimated
Continuation logit scale κ_C	1.254006	[0.02, 50]	Estimated
Owner-service premium χ	1.038033	[0.1, 5]	Estimated
Housing-supply scale H_0	8.162391	[0.2, 80]	Estimated
Bequest intensity θ_0	0.168435	[0, 8]	Estimated
Bequest shift θ_1	0.050046	[0.02, 16]	Estimated; within 2% of lower bound
Childless-renter expenditure share α_0	0.733000	CEX, 2019–2023	Externally fixed
Tenure taste-shock scale	0	Author restriction	Externally fixed
Child dependence of bequest utility	0	Child-invariant bequest	Externally fixed

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